1. Prove that any natural number can be written in binary, i.e., if $n \in \mathbb{N}$, then

$$n = c_k 2^k + c_{k-1} 2^{k-1} + \cdots + c_2 2^2 + c_1 2^1 + c_0 2^0 = (c_k c_{k-1} \ldots c_2 c_1 c_0)_2$$

2. A real number x is said to be algebraic if there exist **integers** $a_0, a_1, \ldots a_n$ such that

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0 = 0,$$

that is, it is a root of a polynomial with integer coefficients. Prove that the set of algebraic numbers is countable.

3. Suppose that A and B are sets bounded above. Prove that if $A \subseteq B$, then $\sup(A) \leq \sup(B)$.

4. Define $A + B = \{a + b : a \in A, b \in B\}$.

   a) Determine $\{1, 3, 5\} + \{-3, 0, 1\}$.

   b) Prove that $\sup(A + B) = \sup(A) + \sup(B)$.

5. Types of definitions/results/theorems that one usually expect a 311 student to know at this point
   (a) Convergent sequence
   (b) Cauchy sequence
   (c) Neighborhood of a real number x
   (d) Accumulation point of a set $S \subset \mathbb{R}$
   (e) Bolzano-Weierstrass Theorem
   (f) Theorems involving sequences $(a_n)_n$ and $(b_n)_n$
   (g) Theorems involving subsequences $(a_{n_k})_k$
   (h) limit of a function

6. Any homework questions?